Pradeep pal

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Technical Skills

Languages: C++, JavaScript, SQL
Full-Stack Engineering: React.js, Node.js, Express.js, PostgreSQL, RESTful APIs, Tailwind CSS
Developer Tools: Git/GitHub, Docker, VS Code, PostgreSQL
Core CS Fundamental: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, OOP

Experience

Open Source Contributor & Researcher **Dec. 2025 – Present**
OpenAstronomy | JuliaAstro Ecosystem (GSoC 2026 Aspirant) *Remote*

- **SpectrumBase.jl Infrastructure Proposal:** Authored a 175-hour GSoC proposal titled *Towards SpectrumBase.jl*, designing a unified infrastructure for astrophysical spectroscopy in Julia.
- **High-Performance Computing:** Architected memory-safe, performance-critical middleware systems focused on data synchronization and large-scale scientific data processing.
- **Scientific Middleware Architecture:** Engineered standardized data pipelines to unify spectroscopic workflows across astronomical software tools.
- **Community Alignment & Open Collaboration:** Collaborated with international maintainers to define architectural blueprints, review specs, and align with JuliaAstro organization guidelines.

Projects

Zero Waste Food | *React.js, Node.js, JavaScript, PostgreSQL* | Live Demo **June 2026**

- Developed a full-stack food donation platform connecting Restaurants, Donors, NGOs, and Admins to reduce food waste and improve food distribution.
- Implemented secure role-based authentication, donation management, food claiming, image uploads, expiry tracking, and real-time availability updates.
- Designed responsive dashboards using React.js and integrated REST APIs with PostgreSQL to manage the complete donation workflow.
- Built an intuitive user interface focused on accessibility, responsiveness, and efficient food distribution across multiple user roles.

PTSRIET-PULSE & Academic Portal | *React, Node,Express, PostgreSQL,Tailwind* | Live Demo **2026**

- **Centralized Digital Portal:** Developed a comprehensive academic platform enabling students to access semester-wise curricula, syllabi, and master cheatsheets.
- **Admin Control Center:** Engineered an admin dashboard to dynamically manage course materials, enquiries, events, job postings, and curriculum access locks.

STDAMS — Satellite Threat Detection System | Link | *Python, Data Modeling* **Feb. 2026**

- Developed an advanced monitoring system designed for **Satellite Telemetry & Abnormal Activity Detection** using simulated datasets.
- Implemented pattern recognition algorithms to identify potential orbital threats and unauthorized signal interference.
- Logic recognized at **UHack 4.0 National Hackathon**, securing a Certificate of Merit for technical innovation.

Achievements & Certifications

- **UHack 4.0 National Hackathon** | Certificate of Merit for Security Prototype. **Feb. 2026**
- **LeetCode Programming** | Solved 20+ problems in C++ (Strong focus on Graph/DP). **2025**

Education

Prof. Rajendra Singh (Rajju Bhaiya) University — Prayagraj, UP **Sep. 2024 – present**
Bachelor of Computer Applications (BCA) *Uttar Pradesh*